DATABASE MANAGEMENT SYSTEM

SQL QUERIES

ASSIGNMENT- 3

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Write an SQL query to display all student records from the Student table.

```
create table sam(studentID INT,name VARCHAR(10),dept VARCHAR(10),marks INT,city VARCHAR(20));
INSERT INTO sam VALUES(1,'VARSHITHA','CSE',90,'CHENNAI');
INSERT INTO sam VALUES(2,'VIKASHINI','ECE',98,'ERODE');
INSERT INTO sam VALUES(3,'DEEPAK','EEE',89,'MADURAI');
INSERT INTO sam VALUES(4,'SANDYA','AIDS',60,'ERODE');
INSERT INTO sam VALUES(5,'ZULFIYA','CSE',50,'NULL');
select*from sam;
```

Output

studentID	name	dept	marks	city
1	VARSHITHA	CSE	90	CHENNAI
2	VIKASHINI	ECE	98	ERODE
3	DEEPAK	EEE	89	MADURAI
4	SANDYA	AIDS	60	ERODE
5	ZULFIYA	CSE	50	NULL

Write a query to display the Name and Marks of all students.

```
SELECT name,marks FROM sam;
```

Output

name	marks
VARSHITHA	90
VIKASHINI	98
DEEPAK	89
SANDYA	60
ZULFIYA	50

Write a query to display students who scored more than 90 marks.

```
SELECT *from sam where marks>90;
```

Output

studentID	name	dept	marks	city
2	VIKASHINI	ECE	98	ERODE

Write a query to display students belonging to the ECE department.

```
SELECT *from sam where dept='ECE';
```

Output

studentID	name	dept	marks	city
2	VIKASHINI	ECE	98	ERODE

Write a query to display students whose city is Chennai.

```
SELECT *from sam where city='CHENNAI';
```

Output

studentID	name	dept	marks	city
1	VARSHITHA	CSE	90	CHENNAI

Write a query to display students whose marks are between 50 and 80.

```
SELECT *from sam where marks between 50 and 80;
```

Output

studentID	name	dept	marks	city
4	SANDYA	AIDS	60	ERODE
5	ZULFIYA	CSE	50	NULL

Write a query to display student names that end with the letter 'a' using the LIKE operator.

```
SELECT name from sam where name like'%a';
```

Output

name
VARSHITHA
SANDYA
ZULFIYA

Write a query to display distinct departments from the Student table.

```
SELECT distinct dept from sam;
```

Output

dept
CSE
ECE
EEE
AIDS

Write a query to display students whose City value is NULL.

```
select * from sam where city is NULL;
```

Output

SQL query successfully executed. However, the result set is empty.

Write a query to find the highest marks obtained by a student using an aggregate function.

```
select max(marks) from sam;
```

Output

max(marks)
98

Write a query to find the average marks of students in the

CSE department.

```sql
select avg(marks)from sam where dept='CSE';
```

Output

avg(marks)
70

Write a query to count the number of students in each department using GROUP BY.

```
select dept,count(*) from sam group by dept;
```

Output

dept	count(*)
AIDS	1
CSE	2
ECE	1
EEE	1

Write a query to display students whose marks are greater than the average marks of all students (nested subquery).

```
select*from sam where marks>(select avg (marks) from sam);
```

Output

studentID	name	dept	marks	city
1	VARSHITHA	CSE	90	CHENNAI
2	VIKASHINI	ECE	98	ERODE
3	DEEPAK	EEE	89	MADURAI

Write an SQL statement to insert a new student record into the Student table.

```
insert into sam values(6,'abi','IT',85,'salam');
select*from sam;
```

Output

studentID	name	dept	marks	city
1	VARSHITHA	CSE	90	CHENNAI
2	VIKASHINI	ECE	98	ERODE
3	DEEPAK	EEE	89	MADURAI
4	SANDYA	AIDS	60	ERODE
5	ZULFIYA	CSE	50	NULL
6	abi	IT	85	salam